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HUB MOTOR WIRE OUTLET STRUCTURE

Abstract

A hub motor wire outlet structure includes a hub motor and a wire outlet assembly. The wire outlet assembly includes a motor connector and a wire connector, and a load-side circuit board of the motor connector and a power-side circuit board of the wire connector are interconnectable through a plurality of terminal pin connecting pillars so as to form a modularized structure. As such, the firmness and alignment accuracy of the assembly can be effectively improved, making it easy to assemble during mounting, enhance operation safety, and also reduce the area occupied by the entire wire assembly.

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Background/Summary

BACKGROUND OF THE INVENTION

(a) Technical Field of the Invention

[0001] The present invention relates to the technical field of hubs of electric vehicles, and more particularly to a modularized hub motor wire outlet structure, by which firmness and alignment accuracy of assembly can be improved and the waterproof and dustproof effect can be enhanced to thereby effectively improve the operation safety thereof.

(b) Description of the Prior Art

[0002] Electrically-assisted bicycles that use hub motors to assist the driving force generated by human power are now widely used. Since the signal lines used for control and the power lines used for power supply in the hub motors need to be connected to the electric machine located inside the hub motor from the control system and the power system on the vehicle frame located on the outside, the common practice is to arrange the wire assembly of the signal line and the power line to individually extend through the hub axle to connect to the power line and the signal line in series with the electric machine. Therefore, in the arrangement operation of the wire assembly, it is not only difficult to assemble, but also has poor fixation and inaccurate alignment. Moreover, it is impossible to effectively perform waterproof and dustproof operations. Moreover, after the overall installation, the signal line and the power line need to be connected one by one. Considering the assembly operation space and convenience, the area occupied by the entire wire assembly is increased, and even causes the problem of easy damage and short service life, and this seriously affects the safety of subsequent riding of the electrically-assisted bicycle.

[0003] In other words, the existing wire assembly structure design for hub motors is not perfect, resulting in the difficulty of wire outgoing assembly operation, large space and insufficient safety. How to solve the above problems is what the present invention intends to explore.

SUMMARY OF THE INVENTION

[0004] One of the primary objectives of the present invention is to make wire outlet connection in a modularized structure so as to improve firmness and alignment accuracy of the assembly to thereby ease assembling during mounting.

[0005] Another one of the primary objectives of the present invention is to effectively enhance the waterproof and dustproof effect so as to effectively improve the operation safety to thereby extend the service life.

[0006] A further one of the primary objectives of the present invention is to reduce the area occupied by a wire assembly so as to enhance convenience of assembling operation and also to reduce the weight of the entire electrically-assisted bicycle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a perspective view of the present invention.

[0008] FIG. **2** is a perspective view of the present invention taken from a different view angle.

[0009] FIG. **3** is a n exploded view of the present invention.

[0010] FIG. **4** is an exploded view of the present invention taken from a different view angle.

[0011] FIG. **5** is a schematic view showing the present invention used in an electrically-assisted bicycle.

[0012] FIG. **6** is a cross-sectional view showing the present invention used in an electrically-assisted bicycle.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0013] The present invention is applicable to driving of an electrically assisted bicycle or an electric vehicle, and, as shown in FIGS. **1**, **2**, and **5**, comprises a hub motor **60** mounted on a vehicle frame **80** of the electrically assisted bicycle or the electric vehicle and a wire outlet assembly **10**, wherein the hub motor **60** includes, arranged at a center thereof, a hub axle **66**, and the hub motor **60** includes an internal cap **61** that is in the form of a cup or bowl. The internal cap **61** includes a motor assembly **62** arranged in an interior thereof to generate an acting force for driving a hub housing **50** (see FIGS. **5** and **6**) to rotate relative to the hub axle **66**. The internal cap **61** is provided with a mounting trough **63** formed in an outside of a closed end thereof and concentrical with the hub axle **66**, and the mounting trough **63** includes at least one first positioning portion **64**. The first positioning portion **64** can be in the form of a hole or in the form of a peg, and in the present invention, the first positioning portion **64** is made in the form of two holes as a primary embodiment for mounting and positioning the wire outlet assembly **10**. Further, the internal cap **61** includes at least one wire outgoing aperture **65** for at least one wire of the motor assembly **62** of the hub motor **60** to extend out therethrough, wherein the wire can be a signal line for control purposes or a power line for power supply purposes. Two ends of the hub axle **66** are each formed with at least one first constraining portion **660** for fixing with a fastening plate **67** having a second constraining hole **68** in mounting the hub motor **60** to the vehicle frame **80** through the hub axle **66**.

[0014] The wire outlet assembly **10** comprises a motor connector **20** and a wire connector **30**. Structural details of the wire outlet assembly **10** are as shown in FIGS. **3**, **4**, and **5**, wherein the motor connector **20** is formed, in a center thereof, with an axle hole **21** corresponding to the hub axle **66**. Further, the motor connector **20** is provided, on one side thereof facing the internal cap **61**, with a contact and support plate **22**, and the contact and support plate **22** is formed, in a surface thereof facing the internal cap **61**, with a mounting recess **220** for receiving and mounting a load-side circuit board **25** having a shape corresponding thereto. The contact and support plate **22** is further formed, in the surface thereof facing the internal cap **61**, with a plurality of positioning notches **221** in communication with the mounting recess **220**, and the positioning notches **221** corresponding to the wire outgoing aperture **65** of the internal cap **61**. Further, the motor connector **20** is provided, in a circumferential edge thereof adjacent to the axle hole **21**, with a plurality of terminal pin orifices **23** penetrating two ends thereof. The terminal pin orifices **23** respectively receive a plurality of terminal pin connecting pillars **24** to insert therein, wherein the terminal pin orifices **23** and the terminal pin connecting pillars **24** can be of different diameter sizes for transmission of signals or electrical power. The load-side circuit board **25** includes positioning extension tabs **254** respectively corresponding to the positioning notches **221** of the mounting recess **220** in order to have the load-side circuit board **25** correctly aligned with and positioned in the mounting recess **220** of the motor connector **20**. The contact and support plate **22** of the motor connector **20** is provided, on the surface thereof, with a second positioning portion **222** corresponding to the first positioning portion **64** of the internal cap **61**, and the second positioning portion **222** can be in the form of a peg or in the form of a hole corresponding to the first positioning portion **64**, and in the present invention, the second positioning portion **222** is made in the form of two pegs as a primary embodiment, in order to allow the motor connector **20** to be correctly positioned in the mounting trough **63** of the internal cap **61**. Further, the internal cap **61** is provided with at least one control circuit board **250**. The load-side circuit board **25** is printed with a plurality of load connecting points **251** for connection with wires of the motor assembly **62** or the control circuit board **250** and a plurality of terminal pin connecting points **252** corresponding to the terminal pin orifices **23**, with a conductive connecting line **253** formed between each of the load connecting points **251** and the terminal pin connecting point **252**. Further, a bearing seat **26** extends from one end of the motor connector **20** opposite to the internal cap **61** for mounting a bearing **260**, so that the bearing **260** is used to support the hub housing **50** outside of the hub motor **60** (see FIG. **6**) to allow the hub housing **50** to freely rotate relative to the motor connector **20**. Further, an end surface of the bearing seat **26** of the motor connector **20** is provided with a guide bar **271** and a positioning arc plate **272** for correct alignment of the wire connector **30**. A waterproof ring gasket **28** is interposed and arranged between the bearing seat **26** and the wire connector **30**, and the waterproof ring gasket **28** is provided with a plurality of terminal pin orifices **281**, a guide hole **282**, and a positioning arc slot **283** respectively corresponding to the terminal pin orifices **23**, the guide bar **271**, and the positioning arc plate **272** of the motor connector **20**. Further, a dust protection ring **29** is interposed between the bearing **260** of the bearing seat **26** of the motor connector **20** and the hub housing **50** for protection against water and dust.

[0015] The wire connector **30** is formed, in a center thereof, with an axle hole **31** corresponding to the hub axle **66**. One side of the wire connector **30** corresponding to the motor connector **20** is formed with an assembly trough **32** for enclosing a terminal pin base **35** through injection molding. The wire connector **30** includes a connecting portion **33** for connecting with a wire harness **40**. Further, the terminal pin base **35** is formed, in a center thereof, with an axle hole **350** corresponding to the hub axle **66**. The terminal pin base **35** includes a plurality of terminal pin orifices **351**, a guide hole **352**, and a positioning arc slot **353** formed therein and respectively corresponding to the terminal pin orifices **23**, the guide bar **271**, and the positioning arc plate **272** of the motor connector **20**, so that the wire connector **30** may be accurately connected with the motor connector **20** by means of the terminal pin base **35**. Further, the terminal pin base **35** is formed, on a circumferential edge thereof, with at least one tooth **355**. One side of the terminal pin base **35** corresponding to the wire connector **30** includes a first fitting engagement portion **356** for mounting a power-side circuit board **36**. The power-side circuit board **36** includes a second fitting engagement portion **360** corresponding to the first fitting engagement portion **356** of the terminal pin base **35**, so that the power-side circuit board **36** can be accurately positioned. The power-side circuit board **36** is formed with a plurality of terminal pin orifices **361** corresponding to the terminal pin orifices **351** of the terminal pin base **35** to receive the terminal pin connecting pillars **24** to insert and penetrate therethrough. Ends of the terminal pin connecting pillars **24** are connected with the signal lines or power lines contained in the wire harness **40**.

[0016] As such, a hub motor wire outlet structure that is modularized and is easy to assemble is formed.

[0017] As to an actual operation of the present invention, referring to FIGS. **1-6**, the power-side circuit board **36** is assembled and mounted on the terminal pin base **35**, and the terminal pin connecting pillars **24** that are connected with the wire harness **40** are inserted, one by one, into the power-side circuit board **36** and the terminal pin base **35**. The terminal pin base **35** is then placed in a mold of the wire connector **30** and the injection molding technology is applied to enclose and form the wire connector **30** in which the terminal pin connecting pillars **24** are included.

[0018] Further, the load-side circuit board **25** is assembled and mounted on the motor connector **20**, and the bearing **260** and the dust protection ring **29** are mounted on the bearing seat **26** of the motor connector **20**. Then, the terminal pin connecting pillars **24** of the wire connector **30** are respectively penetrated through the waterproof ring gasket **28** and the motor connector **20**, and opposite ends of the terminal pin connecting pillars **24** are connected to the terminal pin connecting points **252** of the load-side circuit board **25** corresponding thereto so as to have the motor connector **20** assembled and mounted in the mounting trough **63** of the internal cap **61**, and the signal lines and the power lines of the motor assembly **62** received in the internal cap **61** are connected through the control circuit board **250** to the load connecting point **251** of the load-side circuit board **25** corresponding thereto. Finally, as shown in FIGS. **5** and **6**, the hub housing **50** is mounted outside the hub motor **60**, and stays on two side of the vehicle frame **80** are mounted to the hub axle **66** of the hub motor **60**, so that the wire outlet assembly **10** can be accurately and quickly assembled and mounted to the hub motor **60** in a modularized structure.

[0019] Through the above-mentioned design and description, the present invention can use the plurality of terminal pin connecting pillars **24** to connect the load-side circuit board **25** of the motor connector **20** and the power-side circuit board **36** of the wire connector **30** in the wire outlet assembly **10** to form a modularized structure. This can effectively improve the firmness and alignment accuracy of the assembly, making it easy to assemble during mounting, and can also effectively improve the waterproof and dustproof effect thereof so as to effectively enhance its safety in use and further extend its service life. At the same time, the area occupied by the entire wire assembly can be reduced to further enhance the convenience of assembly operation, and also reduce the weight of the entire electrically-assisted bicycle.

Claims

1. A hub motor wire outlet structure, which is applicable to a vehicle frame of an electrically assisted bicycle, the hub motor wire outlet structure comprising: a hub motor, which comprises a hub axle arranged on a center thereof, the hub motor comprising an internal cap through which a motor assembly is arranged on the hub axle to generate an acting force for driving a hub housing to rotate relative to the hub axle, the internal cap having a closed end that is formed, in an outside thereof, with a mounting trough concentrical with the hub axle, the mounting trough comprising at least one first positioning portion, the internal cap comprising at least one wire outgoing aperture for at least one wire of the motor assembly of the hub motor to extend out therethrough; and a wire outlet assembly, which extends through the hub axle and is arranged in the mounting trough of the internal cap, the wire outlet assembly comprising a motor connector and a wire connector; wherein the motor connector comprises an axle hole formed in a center thereof and corresponding to the hub axle, the motor connector comprising a contact and support plate arranged on one side thereof corresponding to the internal cap, the contact and support plate being formed with a mounting recess in a surface thereof facing the internal cap for receiving and mounting a load-side circuit board having a shape corresponding thereto, the motor connector being provided, in a circumferential edge thereof adjacent to the axle hole, with a plurality of terminal pin orifices penetrating two ends thereof, the terminal pin orifices respectively receiving a plurality of terminal pin connecting pillars to insert therein, the contact and support plate of the motor connector comprising a second positioning portion arranged on a surface thereof and corresponding to the first positioning portion, the load-side circuit board being printed with a plurality of load connecting points connected with the wire of the motor assembly and a plurality of terminal pin connecting points corresponding to the terminal pin orifices, a conductive connecting line being formed between each of the load connecting points and a corresponding one of the terminal pin connecting points, a guide bar and a positioning arc plate being arranged on an end surface of the motor connector for correct alignment of the wire connector; and the wire connector comprises an axle hole formed in a center thereof and corresponding to the hub axle, the wire connector comprising an assembly trough formed in one side thereof corresponding to the motor connector for receiving and mounting a terminal pin base, the wire connector comprising a connecting portion connected to a wire harness, the terminal pin base comprising an axle hole formed in a center thereof and corresponding to the hub axle, the terminal pin base comprising a plurality of terminal pin orifices, a guide hole, and a positioning arc slot formed therein and respectively corresponding to the terminal pin orifices, the guide bar, and the positioning arc plate of the motor connector, the terminal pin base comprising a first fitting engagement portion arranged on one side thereof corresponding to the wire connector for mounting a power-side circuit board, the power-side circuit board comprising a second fitting engagement portion corresponding to the first fitting engagement portion, the power-side circuit board being formed with a plurality of terminal pin orifices corresponding to the terminal pin orifices of the terminal pin base to receive the terminal pin connecting pillars to penetrate therethrough, ends of the terminal pin connecting pillars being connected to the wire of the wire harness.

2. The hub motor wire outlet structure according to claim 1, wherein the first positioning portion of

the internal cap is in the form of a hole, and the second positioning portion of the motor connector is in the form of a peg corresponding to the first positioning portion so as to allow the motor connector to be accurately positioned in the mounting trough of the internal cap.
3. The hub motor wire outlet structure according to claim 1, wherein the surface of the contact and support plate of the motor connector is formed with a plurality of positioning notches in communication with the mounting recess, the positioning notches corresponding to the wire outgoing aperture of the internal cap, the load-side circuit board comprising positioning extension tabs respectively corresponding to the positioning notches of the mounting recess in order to allow the load-side circuit board to be accurately aligned with and positioned in the mounting recess of the motor connector.
4. The hub motor wire outlet structure according to claim 1, wherein the internal cap is provided with at least one control circuit board, the load-side circuit board being printed with a load connecting point connected with a wire of the control circuit board.
5. The hub motor wire outlet structure according to claim 1, wherein a bearing seat extends from one end of the motor connector that is opposite to the internal cap for mounting a bearing, so that the bearing supports the hub housing outside of the hub motor.
6. The hub motor wire outlet structure according to claim 5, wherein a waterproof ring gasket is interposed between the bearing seat of the motor connector and the wire connector, and the waterproof ring gasket is formed with a plurality of terminal pin orifices, a guide hole, and a positioning arc slot respectively corresponding to the terminal pin orifices, the guide bar, and the positioning arc plate of the motor connector.
7. The hub motor wire outlet structure according to claim 5, wherein a dust protection ring is interposed between the bearing of the bearing seat of the motor connector and the hub housing for protection against water and dust.
8. The hub motor wire outlet structure according to claim 1, wherein at least one tooth is formed on a circumferential edge of the terminal pin base, so that the terminal pin base is enclosed in the wire connector through injection molding.
9. The hub motor wire outlet structure according to claim 1, wherein two ends of the hub axle are each formed with at least one first constraining portion for fixing with a fastening plate having a second constraining hole in mounting the hub motor to the vehicle frame through the hub axle.